WEB DEVELOPMENT CASE STUDIES

Problem Statement • Objectives • Algorithm • Flowchart • Pseudocode • HTML/CSS Code • Output Screenshots

CASE STUDY 1: E-Learning Website – LearnSphere

1. Problem Statement

To design and develop a responsive E-Learning Website named LearnSphere that provides students with an easy and interactive platform to explore online courses. The website should display course categories, popular courses, instructor profiles, and relevant information through a user-friendly interface. It should include a navigation bar, attractive layouts, and organized course details to help students find suitable learning resources easily. The website should be developed using HTML and CSS and should work properly on desktop and mobile devices.

2. Objectives

1. To provide a navigation bar for easy access to different sections of the website.
2. To display various online course categories for students.
3. To showcase popular courses with course details and images.
4. To display instructor profiles and their information.
5. To organize learning resources in an attractive and structured layout.
6. To ensure the website works properly on desktop and mobile devices.
7. To improve the user experience through a simple and attractive design.
8. To develop a responsive and user-friendly e-learning website using HTML and CSS.

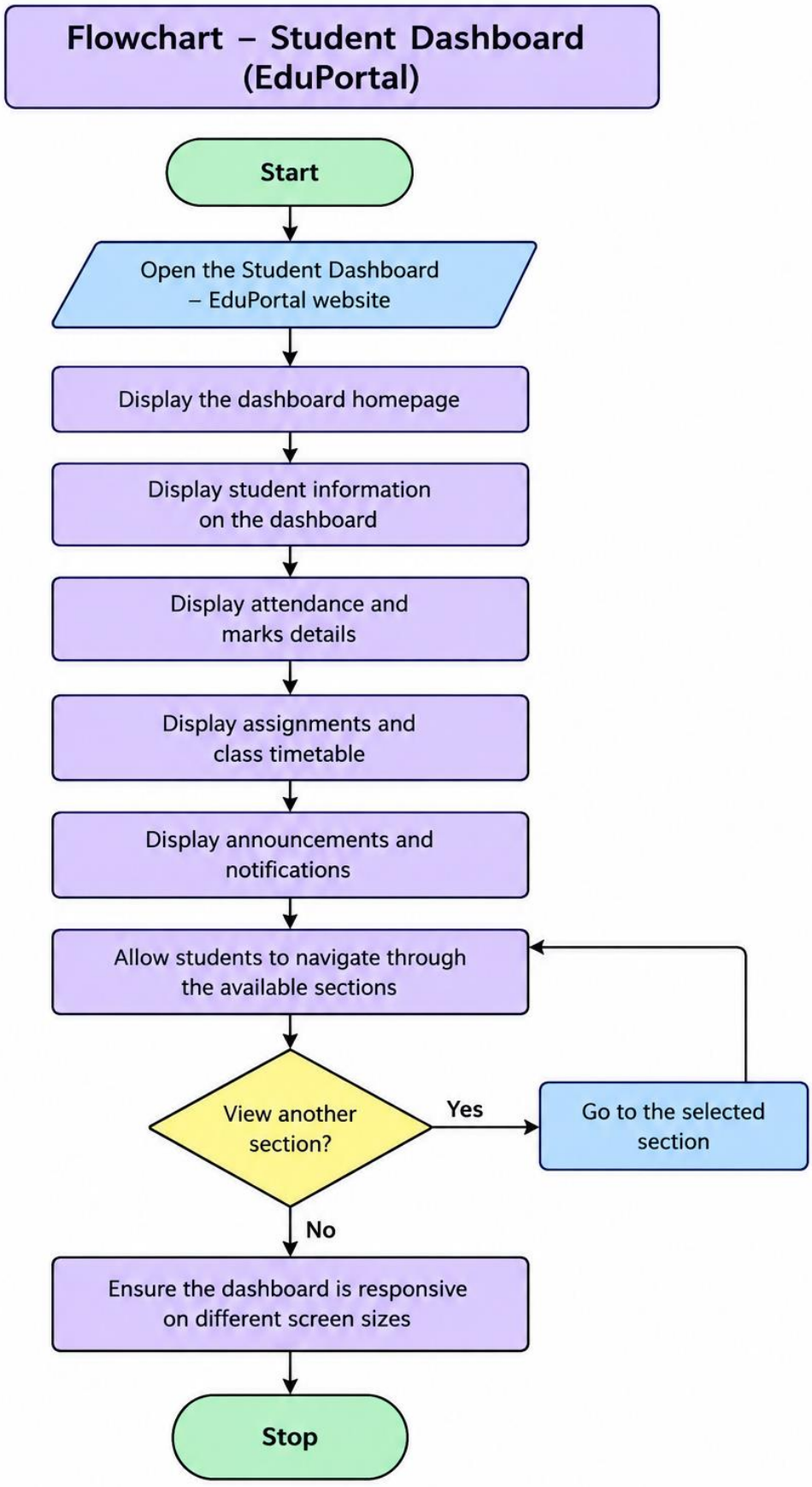
3. Assumptions

1. It is assumed that users have basic knowledge of using a website.
2. The website is designed mainly for students who want to explore online courses.
3. It is assumed that course details, images, and instructor information are available.
4. The website uses HTML to create the structure and CSS to design the layout.
5. The website displays course categories, popular courses, and instructor profiles.
6. It is assumed that users have a device with a web browser to access the website.
7. The website is mainly a front-end design and does not require a database or user login system.
8. The website is designed to be responsive on desktop and mobile devices

4. Algorithm

1. Start
2. Create the basic structure of the website using HTML.
3. Design the navigation bar with Home, Courses, About, and Contact sections.
4. Display the available course categories.
5. Add popular courses with their names, images, and details.
6. Display instructor profiles and their information.
7. Apply CSS styles, colors, fonts, spacing, and layouts.
8. Make the website responsive for desktop and mobile devices.
9. Test all sections to ensure the website displays correctly.
10. Stop

5. Flowchart


Flowchart – Student Dashboard (EduPortal)
Start
Open the Student Dashboard – EduPortal website
Display the dashboard homepage
Display student information on the dashboard
Display attendance and marks details
Display assignments and class timetable
Display announcements and notifications
Allow students to navigate through the available sections
View another section?
Yes
Go to the selected section
No
Ensure the dashboard is responsive on different screen sizes
Stop

6. pseudocode

```
BEGIN
    CREATE LearnSphere webpage
    DISPLAY navigation and home banner
    DISPLAY course categories
    DISPLAY popular course cards
    DISPLAY instructor profiles
    DISPLAY footer information
    APPLY responsive CSS
    TEST page layout
END
```

7.HTML/CSS Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>LearnSphere - E-Learning Website</title>

    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
```

```
body {
    font-family: Arial, sans-serif;
    background: #f4f7fc;
    color: #222;
    line-height: 1.6;
}

header {
    background: #142850;
    color: white;
    padding: 18px 7%;
    display: flex;
    justify-content: space-between;
    align-items: center;
    flex-wrap: wrap;
    gap: 12px;
}

header h1 {
    color: #00d4ff;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 10px;
    font-weight: bold;
}
```

```
nav a:hover {
    color: #00d4ff;
}

.hero {
    text-align: center;
    padding: 65px 20px;
    background: linear-gradient(120deg, #142850, #27496d);
    color: white;
}

.hero h2 {
    font-size: 36px;
    margin-bottom: 12px;
}

.hero p {
    margin-bottom: 22px;
}

.btn {
    display: inline-block;
    background: #00b8d9;
    color: white;
    padding: 11px 20px;
    border: none;
    border-radius: 6px;
    text-decoration: none;
    cursor: pointer;
```

```
    font-size: 15px;
    font-weight: bold;
}

.btn:hover {
    background: #008eaa;
}

section {
    padding: 40px 7%;
}

.section-title {
    text-align: center;
    margin-bottom: 25px;
    color: #142850;
}

.categories {
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: 12px;
}

.category {
    background: white;
    padding: 14px 24px;
    border-radius: 8px;
```

```
    border: 1px solid #dce4f0;
    color: #142850;
    font-weight: bold;
}

.course-container {
    display: grid;
    grid-template-columns:
        repeat(auto-fit, minmax(260px, 1fr));
    gap: 25px;
}

.course-card {
    background: white;
    border-radius: 12px;
    overflow: hidden;
    box-shadow: 0 4px 14px #00000012;
    transition: transform 0.3s;
}

.course-card:hover {
    transform: translateY(-5px);
}

.thumbnail {
    height: 170px;
    position: relative;
    display: flex;
    justify-content: center;
```

```
    align-items: center;
    font-size: 65px;
    color: white;
}

.python { background: linear-gradient(135deg, #306998, #ffd43b); }
.web { background: linear-gradient(135deg, #e44d26, #264de4); }
.ai { background: linear-gradient(135deg, #5426a8, #00b8d9); }

.play {
    position: absolute;
    right: 15px;
    bottom: 12px;
    background: #ffffffdd;
    color: #142850;
    border-radius: 50%;
    width: 42px;
    height: 42px;
    display: grid;
    place-items: center;
    font-size: 20px;
}

.course-content {
    padding: 20px;
}

.tag {
    display: inline-block;
```

```
    background: #e5f7ff;
    color: #087e9d;
    padding: 4px 9px;
    border-radius: 5px;
    font-size: 12px;
    margin-bottom: 10px;
}

.course-content h3 {
    color: #142850;
    margin-bottom: 8px;
}

.instructor {
    color: #666;
    font-size: 14px;
    margin: 10px 0;
}

.course-content ul {
    padding-left: 20px;
    margin: 12px 0 16px;
    font-size: 14px;
}

.progress-label {
    display: flex;
    justify-content: space-between;
    font-size: 13px;
```

```
    margin: 12px 0 5px;
}

.progress-bar {
    height: 9px;
    background: #e5eaf1;
    border-radius: 10px;
    overflow: hidden;
    margin-bottom: 15px;
}

.progress {
    height: 100%;
    background: #00b894;
    border-radius: 10px;
}

.instructor-container {
    display: grid;
    grid-template-columns:
        repeat(auto-fit, minmax(200px, 1fr));
    gap: 20px;
}

.instructor-card {
    background: white;
    text-align: center;
    padding: 25px;
    border-radius: 10px;
```

```
    box-shadow: 0 4px 14px #00000010;
}

.avatar {
    width: 75px;
    height: 75px;
    background: #d9f4ff;
    border-radius: 50%;
    margin: 0 auto 12px;
    display: grid;
    place-items: center;
    font-size: 35px;
}

.enroll-message {
    color: #087e57;
    font-weight: bold;
    margin-top: 10px;
    font-size: 14px;
}

footer {
    background: #142850;
    color: white;
    text-align: center;
    padding: 22px;
}

@media (max-width: 600px) {
```

```
            header {
                flex-direction: column;
                text-align: center;
            }

            nav a {
                display: inline-block;
                margin: 5px;
                font-size: 14px;
            }

            .hero h2 {
                font-size: 27px;
            }

            section {
                padding: 30px 5%;
            }
        }
    </style>
</head>

<body>

    <header>
        <h1>LearnSphere</h1>
        <nav>
            <a href="#home">Home</a>
            <a href="#categories">Categories</a>
```

```
        <a href="#courses">Courses</a>
        <a href="#instructors">Instructors</a>
    </nav>
</header>

<section class="hero" id="home">
    <h2>Learn New Skills Online</h2>
    <p>Explore expert-led courses and build your future with LearnSphere.</p>
    <a href="#courses" class="btn">Explore Courses</a>
</section>

<section id="categories">
    <h2 class="section-title">Course Categories</h2>
    <div class="categories">
        <div class="category">💻 Web Development</div>
        <div class="category">🐍 Python Programming</div>
        <div class="category">🤖 Artificial Intelligence</div>
        <div class="category">📊 Data Science</div>
    </div>
</section>

<section id="courses">
    <h2 class="section-title">Popular Courses</h2>

    <div class="course-container">

        <!-- Python Course -->
        <div class="course-card">
            <div class="thumbnail python">
```

```
        
        <span class="play">▶</span>
    </div>
    <div class="course-content">
        <span class="tag">Programming</span>
        <h3>Python for Beginners</h3>
        <p>Learn Python basics, variables, loops,
          functions, lists and tuples.</p>
        <p class="instructor">👨‍🏫 Instructor: Rahul Sharma</p>
        <strong>Learning Objectives</strong>
        <ul>
            <li>Understand Python syntax</li>
            <li>Write simple Python programs</li>
            <li>Work with lists and tuples</li>
        </ul>
        <div class="progress-label">
            <span>Course Progress</span>
            <span>60%</span>
        </div>
        <div class="progress-bar">
            <div class="progress" style="width:60%"></div>
        </div>
        <button class="btn"
            onclick="enroll('Python for Beginners')">
            Enroll Now
        </button>
        <p class="enroll-message"></p>
    </div>
</div>
```

```
<!-- Web Development Course -->
<div class="course-card">
  <div class="thumbnail web">
    🌐
    <span class="play">▶</span>
  </div>
  <div class="course-content">
    <span class="tag">Web Design</span>
    <h3>Web Development</h3>
    <p>Build responsive websites using HTML,
      CSS and JavaScript.</p>
    <p class="instructor">👩‍🏫 Instructor: Priya Reddy</p>
    <strong>Learning Objectives</strong>
    <ul>
      <li>Create web pages using HTML</li>
      <li>Style pages with CSS</li>
      <li>Add interactivity with JavaScript</li>
    </ul>
    <div class="progress-label">
      <span>Course Progress</span>
      <span>35%</span>
    </div>
    <div class="progress-bar">
      <div class="progress" style="width:35%"></div>
    </div>
    <button class="btn"
      onclick="enroll('Web Development')">
      Enroll Now
```

```
        </button>
        <p class="enroll-message"></p>
    </div>
</div>

<!-- Artificial Intelligence Course -->
<div class="course-card">
    <div class="thumbnail ai">
        🤘
        <span class="play">▶</span>
    </div>
    <div class="course-content">
        <span class="tag">Artificial Intelligence</span>
        <h3>Introduction to AI</h3>
        <p>Discover artificial intelligence,
          machine learning and real-world applications.</p>
        <p class="instructor">🧑‍🏫 Instructor: Arjun Kumar</p>
        <strong>Learning Objectives</strong>
        <ul>
            <li>Understand AI and machine learning</li>
            <li>Explore AI applications</li>
            <li>Learn basic AI concepts</li>
        </ul>
        <div class="progress-label">
            <span>Course Progress</span>
            <span>20%</span>
        </div>
        <div class="progress-bar">
            <div class="progress" style="width:20%"></div>
```

```
            </div>
            <button class="btn"
               onclick="enroll('Introduction to AI')">
               Enroll Now
            </button>
            <p class="enroll-message"></p>
         </div>
      </div>

   </div>
</section>

<section id="instructors">
   <h2 class="section-title">Meet Our Instructors</h2>

   <div class="instructor-container">
      <div class="instructor-card">
         <div class="avatar">👨‍💻</div>
         <h3>Aswini</h3>
         <p>Python Programming Expert</p>
      </div>

      <div class="instructor-card">
         <div class="avatar">👩‍💻</div>
         <h3>Revathi</h3>
         <p>Web Development Expert</p>
      </div>

      <div class="instructor-card">
```

```
            <div class="avatar">🧑‍🔬</div>
            <h3>Mohanajali</h3>
            <p>Artificial Intelligence Expert</p>
        </div>
    </div>
</section>

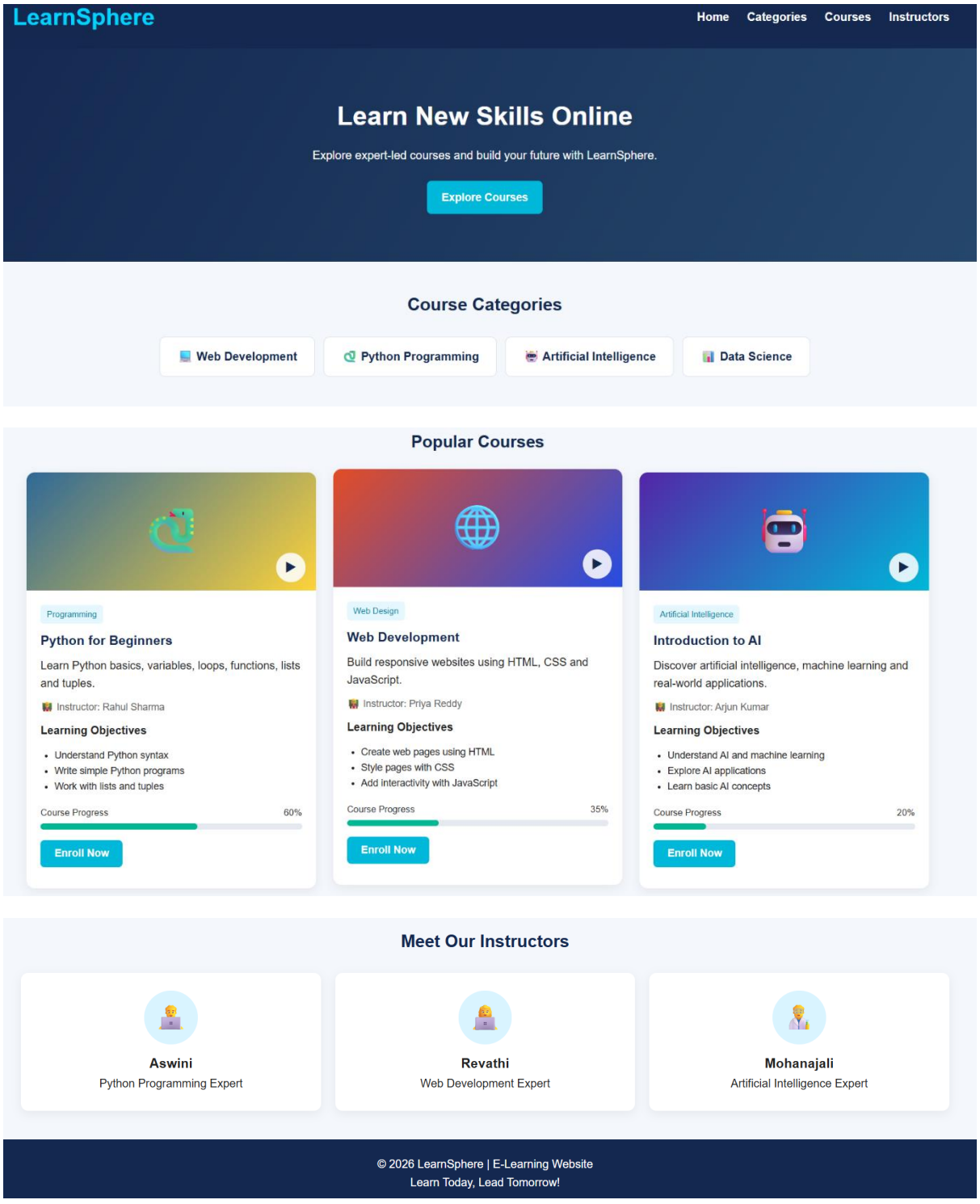
<footer>
    <p>&copy; 2026 LearnSphere | E-Learning Website</p>
    <p>Learn Today, Lead Tomorrow!</p>
</footer>

<script>
    function enroll(courseName) {
        const cards = document.querySelectorAll(".course-card");

        cards.forEach(function(card) {
            const heading = card.querySelector("h3");
            if (heading.textContent === courseName) {
                const message = card.querySelector(".enroll-message");
                message.textContent =
                    "You have selected " + courseName + "!";
            }
        });
    }
</script>

</body>
</html>
```

8.Output Screenshots



CASE STUDY 2 :Student Dashboard – EduPortal

1.Problem Statement

Students often find it difficult to access their academic information, such as attendance, marks, assignments, class schedules, and announcements in one place. Managing these details separately can be time-consuming and may lead to confusion.

The Student Dashboard – EduPortal is designed to provide a simple and user-friendly interface where students can view their academic information in one place. The dashboard displays important details such as student information, attendance, marks, assignments, timetable, and notifications. The main goal is to make academic information easily accessible, organized, and convenient for students.

2.Objectives

The main objectives of the Student Dashboard – EduPortal are:

1. To develop a simple and user-friendly student dashboard.
2. To display student academic information in one place.
3. To provide easy access to attendance details and marks.
4. To display assignments and class schedules clearly.
5. To show important announcements and notifications.
6. To organize student information in a structured manner.
7. To design a responsive interface that works on different screen sizes.
8. To improve the accessibility and usability of academic information.

3.Assumptions

The following assumptions are considered for the Student Dashboard – EduPortal project:

1. Students have access to a computer, laptop, or mobile device.
2. An internet connection is available to access the dashboard.
3. Students have basic knowledge of using a web browser.
4. Student information is assumed to be available for displaying on the dashboard.
5. Attendance, marks, assignments, and timetable details are assumed to be updated correctly.
6. The dashboard is designed to provide a simple and user-friendly interface.
7. The website is expected to work on commonly used web browsers.
8. The dashboard layout is designed to be responsive across different screen sizes.

4.Algorithm

Step 1: Start the process.

Step 2: Open the Student Dashboard – EduPortal website.

Step 3: Display the dashboard homepage.

Step 4: Display student information on the dashboard.

Step 5: Display the attendance and marks details.

Step 6: Display assignments and class timetable.

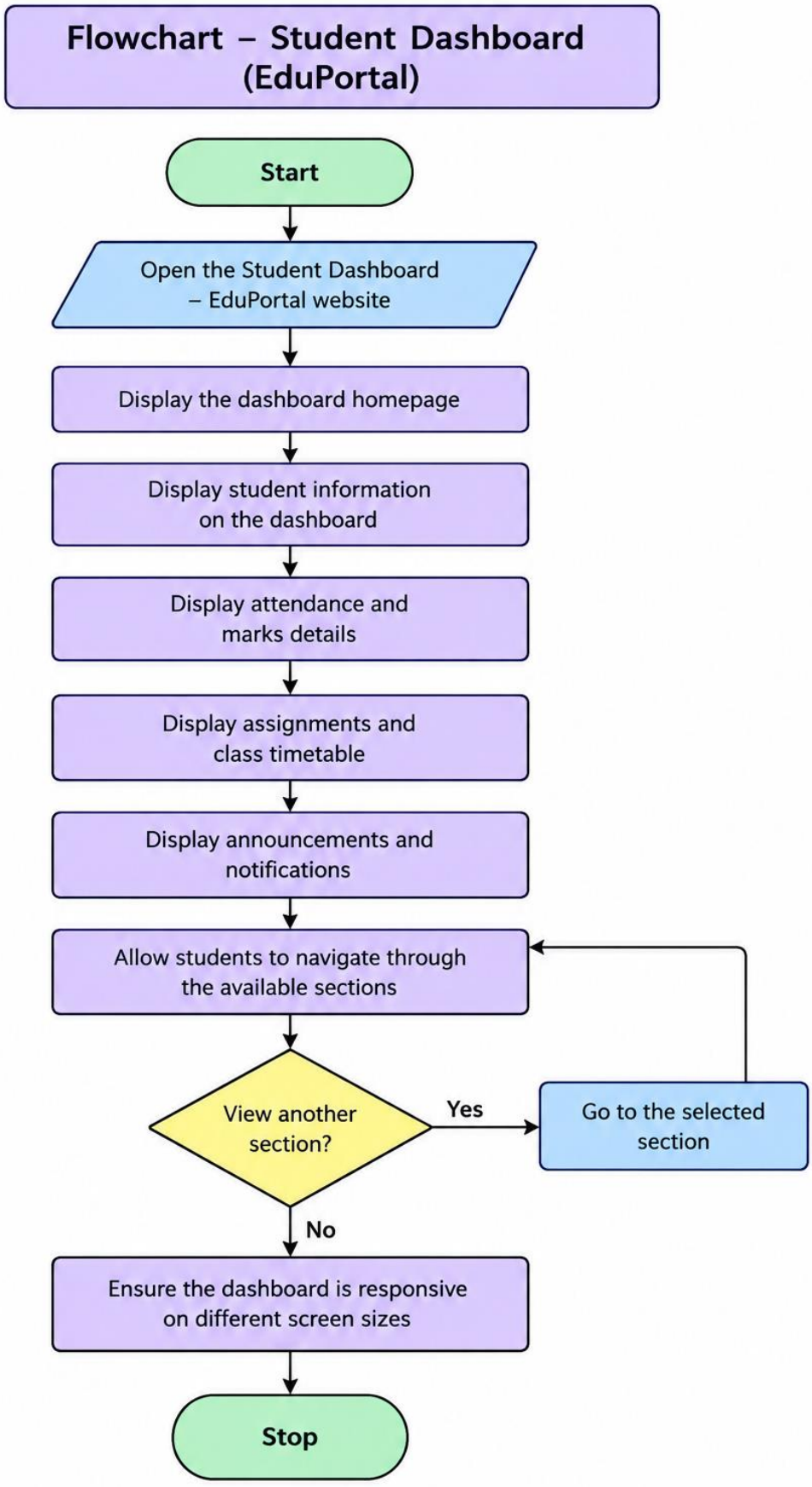
Step 7: Display important announcements and notifications.

Step 8: Allow students to navigate through the available dashboard sections.

Step 9: Ensure that the dashboard layout is displayed properly on different screen sizes.

Step 10: Stop the process.

5.Flowchart


Flowchart – Student Dashboard (EduPortal)
Start
Open the Student Dashboard – EduPortal website
Display the dashboard homepage
Display student information on the dashboard
Display attendance and marks details
Display assignments and class timetable
Display announcements and notifications
Allow students to navigate through the available sections
View another section?
Yes
Go to the selected section
No
Ensure the dashboard is responsive on different screen sizes
Stop

6.Pseudocode

BEGIN

1. CREATE Student Dashboard – EduPortal webpage.
2. DISPLAY the dashboard homepage.
3. DISPLAY student information.
4. DISPLAY attendance details.
5. DISPLAY marks and academic performance.
6. DISPLAY assignments and class timetable.
7. DISPLAY announcements and notifications.
8. ALLOW students to navigate through the available sections.
9. APPLY responsive CSS for different screen sizes.
10. TEST the webpage layout.

END

7.HTML/CSS Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Dashboard | EduPortal</title>

<style>
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}
```

```
:root {
    --primary: #5146e5;
    --dark: #202447;
    --light: #f5f6fc;
    --text: #323750;
    --muted: #858ba5;
    --border: #e9eaf3;
}

body {
    font-family: "Segoe UI", Arial, sans-serif;
    background: var(--light);
    color: var(--text);
}

button, input {
    font: inherit;
}

.layout {
    display: flex;
    min-height: 100vh;
}

/* SIDEBAR */

.sidebar {
    width: 245px;
    background: var(--dark);
```

```
    color: white;
    padding: 28px 18px;
    position: fixed;
    top: 0;
    bottom: 0;
    left: 0;
    overflow-y: auto;
    z-index: 10;
    transition: transform 0.3s;
}

.logo {
    font-size: 25px;
    font-weight: 800;
    margin: 0 8px 36px;
}

.logo span {
    color: #a9a2ff;
}

.profile {
    text-align: center;
    padding: 18px 10px;
    background: #ffffff0d;
    border-radius: 14px;
    margin-bottom: 28px;
}
```

```
.avatar {
    width: 70px;
    height: 70px;
    border-radius: 50%;
    background: linear-gradient(135deg, #9b8cff, #50d6d0);
    display: grid;
    place-items: center;
    margin: 0 auto 12px;
    color: white;
    font-size: 28px;
    font-weight: bold;
}

.profile h3 {
    font-size: 16px;
}

.profile p {
    color: #b8bbd4;
    font-size: 12px;
    margin-top: 5px;
}

.menu-title {
    color: #8e93b3;
    font-size: 11px;
    letter-spacing: 1.5px;
    margin: 20px 10px 10px;
}
```

```
.nav-link {
    display: flex;
    align-items: center;
    gap: 12px;
    padding: 13px 14px;
    color: #c9cce0;
    text-decoration: none;
    border-radius: 9px;
    margin: 5px 0;
    font-size: 14px;
    transition: 0.2s;
}

.nav-link:hover,
.nav-link.active {
    background: var(--primary);
    color: white;
}

/* MAIN CONTENT */

.main {
    margin-left: 245px;
    width: calc(100% - 245px);
    padding: 28px 32px;
    min-width: 0;
}
```

```
.topbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 16px;
    margin-bottom: 28px;
}

.topbar h1 {
    font-size: 27px;
    color: var(--dark);
}

.topbar p {
    color: var(--muted);
    font-size: 13px;
    margin-top: 5px;
}

.top-actions {
    display: flex;
    align-items: center;
    gap: 10px;
}

.date-badge {
    background: white;
    border: 1px solid var(--border);
    padding: 10px 13px;
```

```
    border-radius: 9px;
    font-size: 13px;
}

.menu-toggle {
    display: none;
    border: none;
    background: var(--primary);
    color: white;
    border-radius: 8px;
    padding: 9px 12px;
    cursor: pointer;
}

.section {
    scroll-margin-top: 20px;
    margin-bottom: 30px;
}

.section-heading {
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 12px;
    margin-bottom: 17px;
}

.section-heading h2 {
    font-size: 19px;
```

```
    color: var(--dark);
}

.section-heading p {
    color: var(--muted);
    font-size: 12px;
}

.grid {
    display: grid;
    grid-template-columns: repeat(4, minmax(0, 1fr));
    gap: 17px;
}

.card {
    background: white;
    border: 1px solid var(--border);
    border-radius: 15px;
    padding: 21px;
    box-shadow: 0 5px 18px #20244705;
}

.stat-card {
    position: relative;
    overflow: hidden;
}

.stat-top {
    display: flex;
```

```
    justify-content: space-between;
    align-items: center;
    margin-bottom: 17px;
}

.stat-icon {
    width: 43px;
    height: 43px;
    display: grid;
    place-items: center;
    border-radius: 12px;
    font-size: 21px;
}

.purple { background: #eeebff; }
.blue { background: #e5f2ff; }
.green { background: #e1f8ef; }
.orange { background: #fff0df; }

.stat-label {
    color: var(--muted);
    font-size: 13px;
}

.stat-value {
    font-size: 28px;
    color: var(--dark);
    font-weight: 800;
}
```

```
.stat-note {
    font-size: 12px;
    color: #18a875;
    margin-top: 7px;
}

.welcome {
    background: linear-gradient(115deg, #5146e5, #8274ff);
    color: white;
    border-radius: 17px;
    padding: 27px 30px;
    margin-bottom: 28px;
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 20px;
}

.welcome h2 {
    font-size: 23px;
    margin-bottom: 8px;
}

.welcome p {
    color: #e9e6ff;
    font-size: 14px;
}
```

```
.welcome-art {
    font-size: 58px;
}

/* PROGRESS BARS */

.progress-item {
    margin: 17px 0;
}

.progress-info {
    display: flex;
    justify-content: space-between;
    gap: 10px;
    font-size: 13px;
    margin-bottom: 9px;
}

.progress-info span:last-child {
    font-weight: bold;
}

.progress-bar {
    height: 9px;
    background: #edf0f7;
    border-radius: 20px;
    overflow: hidden;
}
```

```
.progress-fill {
    height: 100%;
    border-radius: 20px;
}

/* TWO-COLUMN SECTION */

.two-columns {
    display: grid;
    grid-template-columns: minmax(0, 1.5fr) minmax(260px, 1fr);
    gap: 20px;
}

.subject-card {
    display: flex;
    align-items: center;
    gap: 12px;
    margin-top: 17px;
}

.subject-icon {
    width: 43px;
    height: 43px;
    border-radius: 11px;
    display: grid;
    place-items: center;
    font-size: 20px;
    flex-shrink: 0;
}
```

```
.subject-info {
    flex: 1;
    min-width: 0;
}

.subject-info h4 {
    font-size: 13px;
    margin-bottom: 4px;
}

.subject-info p {
    color: var(--muted);
    font-size: 11px;
}

.subject-score {
    font-size: 14px;
    font-weight: 800;
}

/* TABLES */

.table-wrap {
    overflow-x: auto;
}

table {
    border-collapse: collapse;
```

```
    width: 100%;
    min-width: 500px;
    font-size: 13px;
}

th {
    background: #f8f8fd;
    color: #777d98;
    font-size: 11px;
    text-transform: uppercase;
    letter-spacing: 0.4px;
}

th, td {
    padding: 15px 13px;
    text-align: left;
    border-bottom: 1px solid var(--border);
}

tbody tr:last-child td {
    border-bottom: none;
}

tbody tr:hover {
    background: #fafaff;
}

.badge {
    display: inline-block;
```

```
    padding: 5px 9px;
    border-radius: 20px;
    font-size: 11px;
    font-weight: bold;
}

.badge.green-badge {
    background: #e2f8ec;
    color: #158754;
}

.badge.blue-badge {
    background: #e5edff;
    color: #4259c8;
}

.badge.orange-badge {
    background: #fff1dc;
    color: #b67513;
}

/* ANNOUNCEMENTS */

.announcement {
    display: flex;
    gap: 13px;
    padding: 15px 0;
    border-bottom: 1px solid var(--border);
}
```

```
.announcement:last-child {
    border-bottom: none;
}

.announcement-icon {
    min-width: 39px;
    height: 39px;
    display: grid;
    place-items: center;
    border-radius: 10px;
    background: #eeebff;
    font-size: 19px;
}

.announcement h4 {
    font-size: 13px;
    margin-bottom: 5px;
}

.announcement p {
    font-size: 12px;
    color: var(--muted);
}

.announcement small {
    display: block;
    font-size: 10px;
    color: var(--primary);
```

```
    margin-top: 6px;
}

/* FOOTER */

footer {
    text-align: center;
    color: var(--muted);
    font-size: 12px;
    padding: 20px 0 5px;
}

/* RESPONSIVE DESIGN */

@media (max-width: 1100px) {
    .grid {
        grid-template-columns: repeat(2, minmax(0, 1fr));
    }

    .two-columns {
        grid-template-columns: 1fr;
    }
}

@media (max-width: 700px) {
    .sidebar {
        transform: translateX(-100%);
    }
```

```
.sidebar.open {
    transform: translateX(0);
    box-shadow: 10px 0 35px #0003;
}

.main {
    margin-left: 0;
    width: 100%;
    padding: 20px 15px;
}

.menu-toggle {
    display: inline-block;
}

.topbar {
    align-items: flex-start;
}

.topbar h1 {
    font-size: 21px;
}

.date-badge {
    display: none;
}

.welcome {
    padding: 22px;
```

```
    }

    .welcome h2 {
        font-size: 19px;
    }

    .welcome-art {
        font-size: 38px;
    }

    .card {
        padding: 16px;
    }
}

@media (max-width: 400px) {
    .grid {
        grid-template-columns: 1fr;
    }

    .welcome-art {
        display: none;
    }
}
</style>
</head>

<body>
<div class="layout">
```

```
<!-- SIDEBAR -->
<aside class="sidebar" id="sidebar">
    <div class="logo">Edu<span>Portal.</span></div>

    <div class="profile">
        <div class="avatar">VS</div>
        <h3>Varshini Garapati</h3>
        <p>B.Tech | AIML - 1</p>
        <p>Student ID: AIML2026</p>

    </div>

    <p class="menu-title">MAIN MENU</p>

    <a href="#overview" class="nav-link active">🏠 Dashboard</a>
    <a href="#student" class="nav-link">👩‍🎓 Student Details</a>
    <a href="#results" class="nav-link">📊 Semester Results</a>
    <a href="#attendance" class="nav-link">📅 Attendance</a>
    <a href="#subjects" class="nav-link">📚 Subjects</a>
    <a href="#exams" class="nav-link">📝 Exam Schedule</a>
    <a href="#announcements" class="nav-link">🔔 Announcements</a>

    <p class="menu-title">STUDENT PORTAL</p>
    <a href="#student" class="nav-link">⚙️ My Profile</a>
</aside>

<!-- MAIN CONTENT -->
```

```
<main class="main">

    <div class="topbar">
        <div style="display:flex;gap:12px;align-items:flex-start;">
            <button class="menu-toggle" id="menuToggle"
                aria-label="Toggle navigation">☰</button>
            <div>
                <h1>Student Dashboard</h1>
                <p>Welcome back! Here's your academic overview.</p>
            </div>
        </div>

        <div class="top-actions">
            <div class="date-badge">🎓 Academic Year 2026–27</div>
        </div>
    </div>

    <!-- WELCOME -->
    <section class="welcome" id="overview">
        <div>
            <h2>Hello, Varshini! 👏</h2>
            <p>Keep learning, keep growing. Your goals are within reach.</p>
            <p style="margin-top:12px;">B.Tech Artificial Intelligence & Machine
Learning</p>
        </div>
        <div class="welcome-art">🎓</div>
    </section>

    <!-- STUDENT DETAILS -->
```

```
<section class="section" id="student">
    <div class="section-heading">
        <h2>Academic Overview</h2>
        <p>Sample student data</p>
    </div>

    <div class="grid">
        <div class="card stat-card">
            <div class="stat-top">
                <span class="stat-label">Current CGPA</span>
                <div class="stat-icon purple">🏆</div>
            </div>
            <div class="stat-value">9.8</div>
            <p class="stat-note">Semester performance</p>
        </div>

        <div class="card stat-card">
            <div class="stat-top">
                <span class="stat-label">Attendance</span>
                <div class="stat-icon blue">🧮</div>
            </div>
            <div class="stat-value">98%</div>
            <p class="stat-note">Overall attendance</p>
        </div>

        <div class="card stat-card">
            <div class="stat-top">
                <span class="stat-label">Total Subjects</span>
                <div class="stat-icon green">📚</div>
```

```
            </div>
            <div class="stat-value">5</div>
            <p class="stat-note">Current semester</p>
        </div>

        <div class="card stat-card">
            <div class="stat-top">
                <span class="stat-label">Upcoming Exams</span>
                <div class="stat-icon orange">📝</div>
            </div>
            <div class="stat-value">3</div>
            <p class="stat-note" style="color:#b67513;">
                Check exam schedule
            </p>
        </div>
    </div>
</section>

<!-- SEMESTER RESULTS -->
<section class="section" id="results">
    <div class="section-heading">
        <h2>Semester Results</h2>
        <p>Performance summary</p>
    </div>

    <div class="card">
        <div class="table-wrap">
            <table>
                <thead>
```

```
                <tr>
                    <th>Semester</th>
                    <th>Credits</th>
                    <th>SGPA</th>
                    <th>Result</th>
                </tr>
            </thead>
            <tbody>
                <tr>
                    <td>Semester 1</td>
                    <td>99</td>
                    <td>9.8</td>
                    <td><span class="badge green-badge">Pass</span></td>
                </tr>
                <tr>
                    <td>Semester 2</td>
                    <td>--</td>
                    <td>--</td>
                    <td><span class="badge blue-badge">Upcoming</span></td>
                </tr>
            </tbody>
        </table>
    </div>
  </div>
</section>

<!-- ATTENDANCE AND SUBJECTS -->
<div class="two-columns">
```

```
<section class="section" id="attendance">
  <div class="section-heading">
    <h2>Attendance Report</h2>
    <p>Subject-wise</p>
  </div>

  <div class="card">
    <div class="progress-item">
      <div class="progress-info">
        <span>Python Programming</span><span>92%</span>
      </div>
      <div class="progress-bar">
        <div class="progress-fill"
          style="width:92%;background:#5146e5;"></div>
      </div>
    </div>

    <div class="progress-item">
      <div class="progress-info">
        <span>Mathematics</span><span>85%</span>
      </div>
      <div class="progress-bar">
        <div class="progress-fill"
          style="width:95%;background:#24b8a0;"></div>
      </div>
    </div>

    <div class="progress-item">
      <div class="progress-info">
```

```
            <span>Professional Communication</span><span>88%</span>
          </div>
          <div class="progress-bar">
            <div class="progress-fill"
              style="width:98%;background:#f4a340;"></div>
          </div>
        </div>

        <div class="progress-item">
          <div class="progress-info">
            <span>Computational Mathematics</span><span>78%</span>
          </div>
          <div class="progress-bar">
            <div class="progress-fill"
              style="width:98%;background:#5488ef;"></div>
          </div>
        </div>

        <div class="progress-item">
          <div class="progress-info">
            <span>Artificial Intelligence</span><span>90%</span>
          </div>
          <div class="progress-bar">
            <div class="progress-fill"
              style="width:99%;background:#24b8a0;"></div>
          </div>
        </div>
      </div>
    </section>
```

```
<section class="section" id="subjects">
  <div class="section-heading">
    <h2>My Subjects</h2>
    <p>Semester 1</p>
  </div>

  <div class="card">
    <div class="subject-card">
      <div class="subject-icon purple">🐍</div>
      <div class="subject-info">
        <h4>Python Programming</h4>
        <p>CS101 · 4 Credits</p>
      </div>
      <span class="subject-score">A</span>
    </div>

    <div class="subject-card">
      <div class="subject-icon blue">➗</div>
      <div class="subject-info">
        <h4>Computational Mathematics</h4>
        <p>MA101 · 4 Credits</p>
      </div>
      <span class="subject-score">A</span>
    </div>

    <div class="subject-card">
      <div class="subject-icon green">🤖</div>
      <div class="subject-info">
```

```
                <h4>Artificial Intelligence</h4>
                <p>AI101 · 3 Credits</p>
            </div>
            <span class="subject-score">A</span>
        </div>

        <div class="subject-card">
            <div class="subject-icon orange">📹</div>
            <div class="subject-info">
                <h4>Professional Communication</h4>
                <p>EN101 · 2 Credits</p>
            </div>
            <span class="subject-score">A</span>
        </div>

        <div class="subject-card">
            <div class="subject-icon purple">💻</div>
            <div class="subject-info">
                <h4>Web Development</h4>
                <p>WD101 · 3 Credits</p>
            </div>
            <span class="subject-score">A</span>
        </div>
    </div>
  </section>

</div>

<!-- EXAMINATION SCHEDULE -->
```

```
<section class="section" id="exams">
    <div class="section-heading">
        <h2>Examination Schedule</h2>
        <p>Upcoming examinations</p>
    </div>

    <div class="card">
        <div class="table-wrap">
            <table>
                <thead>
                    <tr>
                        <th>Subject</th>
                        <th>Date</th>
                        <th>Time</th>
                        <th>Exam Type</th>
                        <th>Status</th>
                    </tr>
                </thead>
                <tbody>
                    <tr>
                        <td>Python Programming</td>
                        <td>10 Nov 2026</td>
                        <td>9:30 AM</td>
                        <td>Mid Examination</td>
                        <td><span class="badge orange-
badge">Upcoming</span></td>
                    </tr>
                    <tr>
                        <td>Computational Mathematics</td>
```

```
                        <td>12 Nov 2026</td>
                        <td>9:30 AM</td>
                        <td>Mid Examination</td>
                        <td><span class="badge orange-
badge">Upcoming</span></td>
                    </tr>
                    <tr>
                        <td>Professional Communication</td>
                        <td>14 Nov 2026</td>
                        <td>1:30 PM</td>
                        <td>Mid Examination</td>
                        <td><span class="badge blue-badge">Scheduled</span></td>
                    </tr>
                </tbody>
            </table>
        </div>
    </div>
</section>

<!-- ANNOUNCEMENTS -->
<section class="section" id="announcements">
    <div class="section-heading">
        <h2>Announcements & Notices</h2>
        <p>Latest updates</p>
    </div>

    <div class="card">
        <div class="announcement">
            <div class="announcement-icon">📢</div>
```

```
            <div>
                <h4>Mid Examination Schedule</h4>
                <p>Students should check the examination timetable
                  and prepare accordingly.</p>
                <small>Academic Notice · 24 Oct 2026</small>
            </div>
        </div>

        <div class="announcement">
            <div class="announcement-icon">📚</div>
            <div>
                <h4>Assignment Submission</h4>
                <p>Submit pending programming assignments before
                  the announced deadline.</p>
                <small>Department Notice · 23 Oct 2026</small>
            </div>
        </div>

        <div class="announcement">
            <div class="announcement-icon">🎓</div>
            <div>
                <h4>Attendance Reminder</h4>
                <p>Regular class attendance is important.
                  Check your subject-wise attendance report.</p>
                <small>Student Notice · 22 Oct 2026</small>
            </div>
        </div>
    </div>
</section>
```

```
        <footer>
            © 2026 EduPortal Student Dashboard | Designed for Students
        </footer>

    </main>
</div>

<script>
    // Mobile sidebar toggle
    const sidebar = document.getElementById("sidebar");
    const menuToggle = document.getElementById("menuToggle");

    menuToggle.addEventListener("click", function() {
        sidebar.classList.toggle("open");
    });

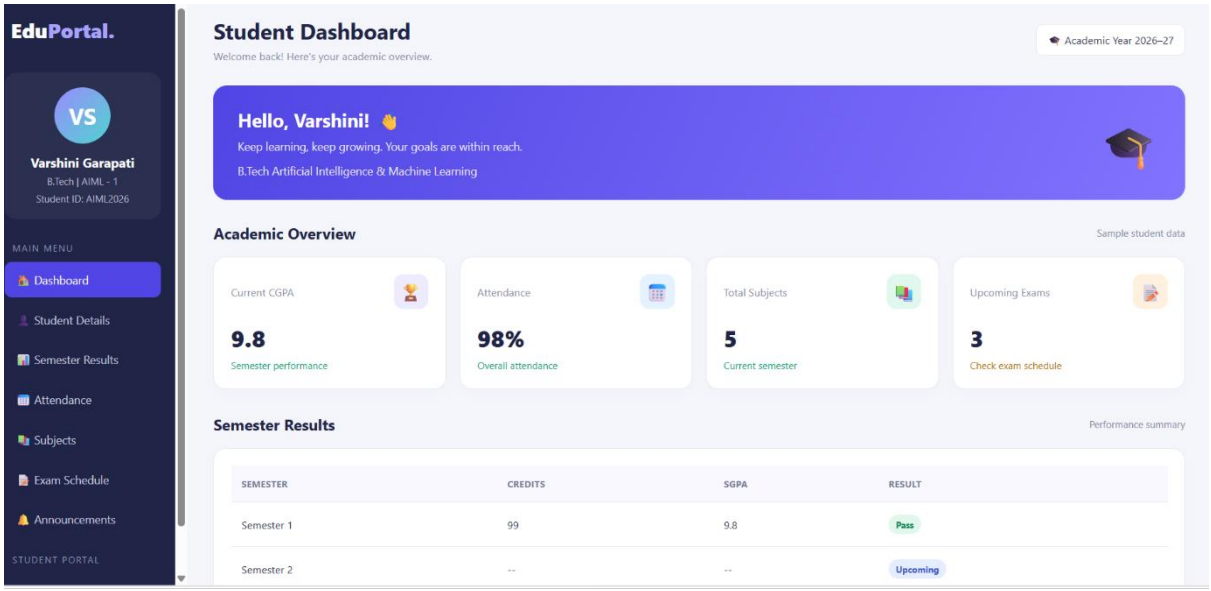
    // Highlight active navigation link and close mobile menu
    document.querySelectorAll(".nav-link").forEach(function(link) {
        link.addEventListener("click", function() {
            document.querySelectorAll(".nav-link").forEach(function(item) {
                item.classList.remove("active");
            });

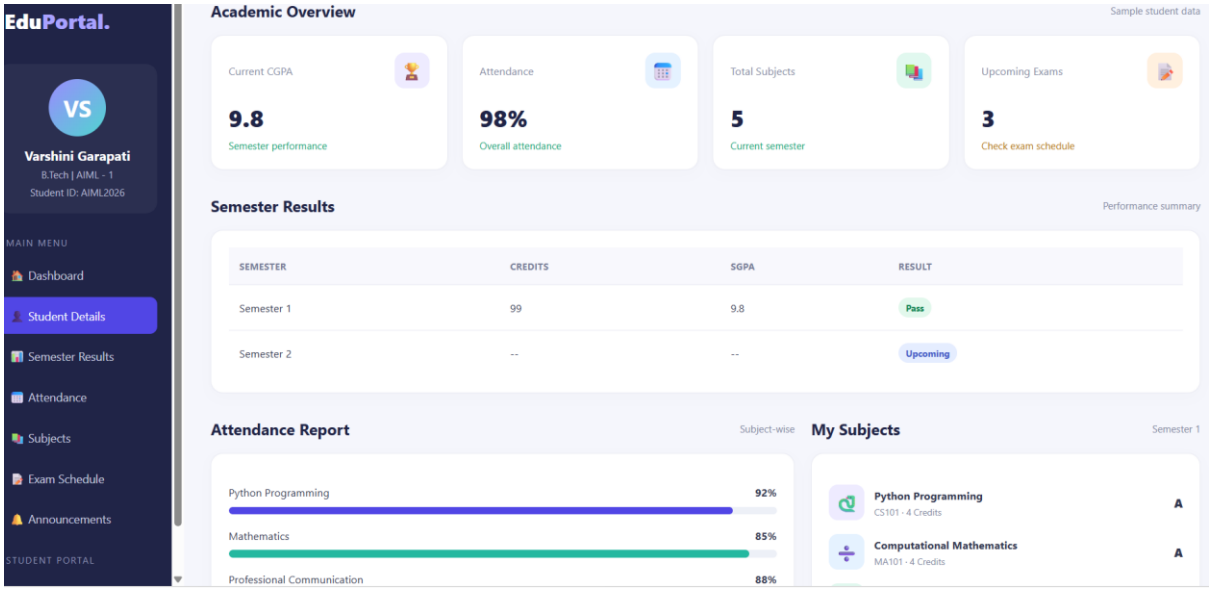
            this.classList.add("active");
            sidebar.classList.remove("open");
        });
    });
</script>
```

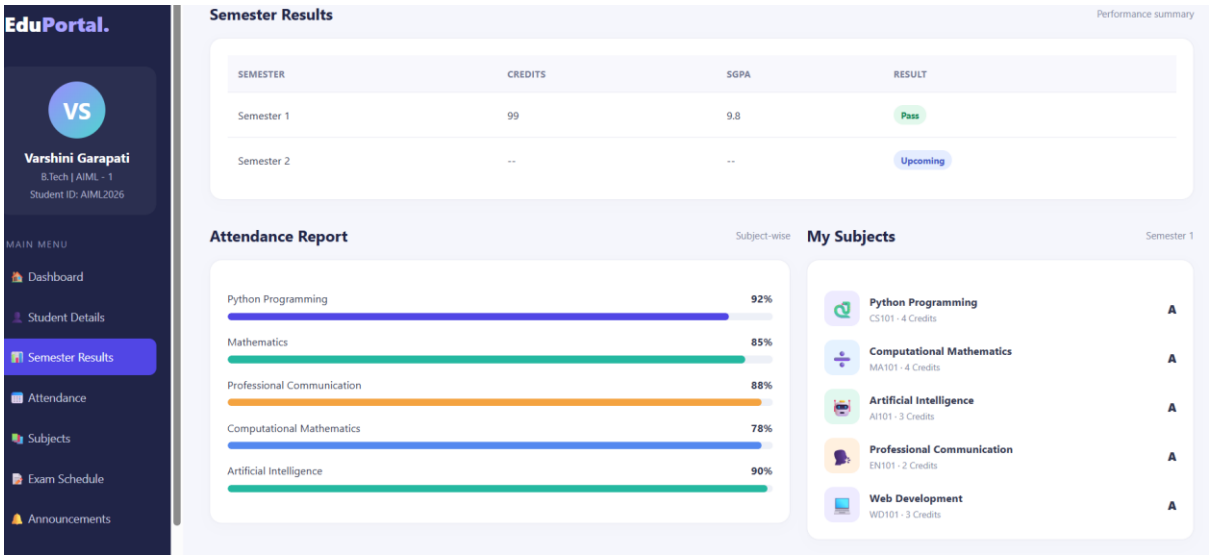
</body>

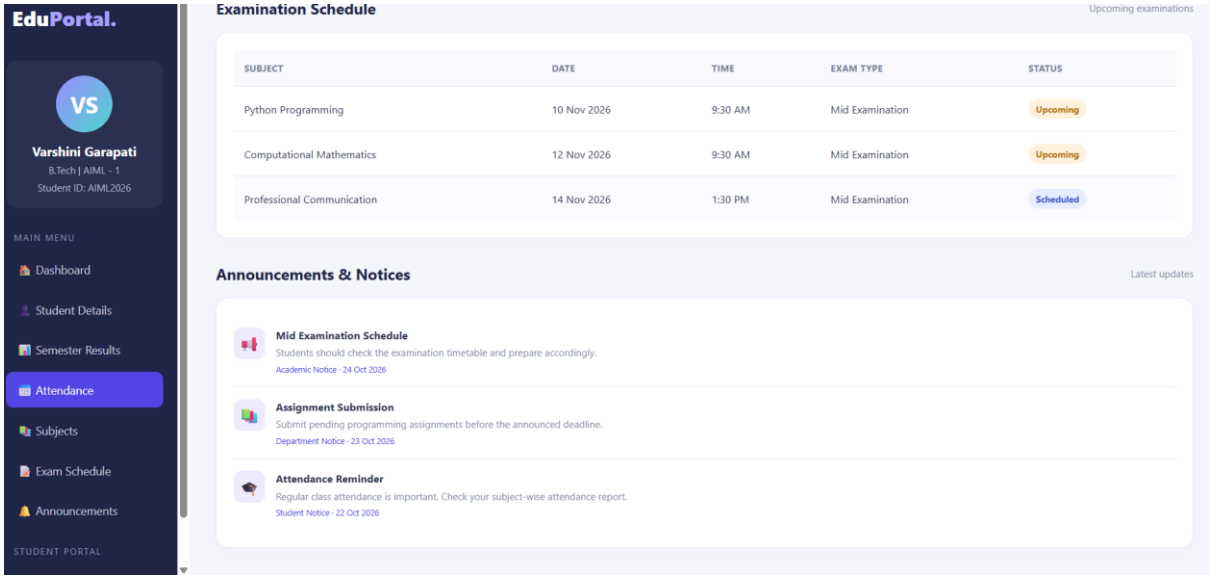
</html>

8.Output Screenshots









CASE STUDY 3:Doctor Appointment Website

1.Problem Statement

Patients often face difficulties in finding suitable doctors, checking their availability, and scheduling appointments. Booking appointments through phone calls or visiting hospitals in person can be time-consuming and inconvenient.

The Doctor Appointment Website is designed to provide a simple and user-friendly platform where patients can view doctor details, check available appointment information, and book appointments easily. The website aims to reduce the difficulties involved in appointment scheduling and make the process more convenient for both patients and doctors.

2. Objectives

The main objectives of the Doctor Appointment Website are:

1. To develop a simple and user-friendly doctor appointment website.
2. To display doctor details, including names and specializations.
3. To help patients find suitable doctors easily.
4. To provide information about doctors and their availability.
5. To make the appointment booking process simple and convenient.
6. To reduce the time required to schedule doctor appointments.
7. To design a responsive website that works on different screen sizes.
8. To improve the accessibility of doctor and appointment information.

3.Assumptions

The following assumptions are considered for the Doctor Appointment Website project:

1. Users have access to a computer, laptop, or mobile device.
2. An internet connection is available to access the website.
3. Users have basic knowledge of using a web browser.
4. Doctor details, including names and specializations, are assumed to be available on the website.
5. Appointment availability information is assumed to be displayed correctly.
6. Users can easily navigate through the website to find doctor information.
7. The website is expected to work on commonly used web browsers.
8. The website layout is designed to be responsive across different screen sizes.
9. The website provides a simple and user-friendly interface.

10. Users are expected to enter correct information when using the appointment form, if available.

4.Algorithm

Step 1: Start the process.

Step 2: Open the Doctor Appointment Website.

Step 3: Display the homepage.

Step 4: Display the available doctors and their specializations.

Step 5: Allow users to view doctor details.

Step 6: Display appointment availability information.

Step 7: Allow users to select a doctor and enter appointment details, if a booking form is available.

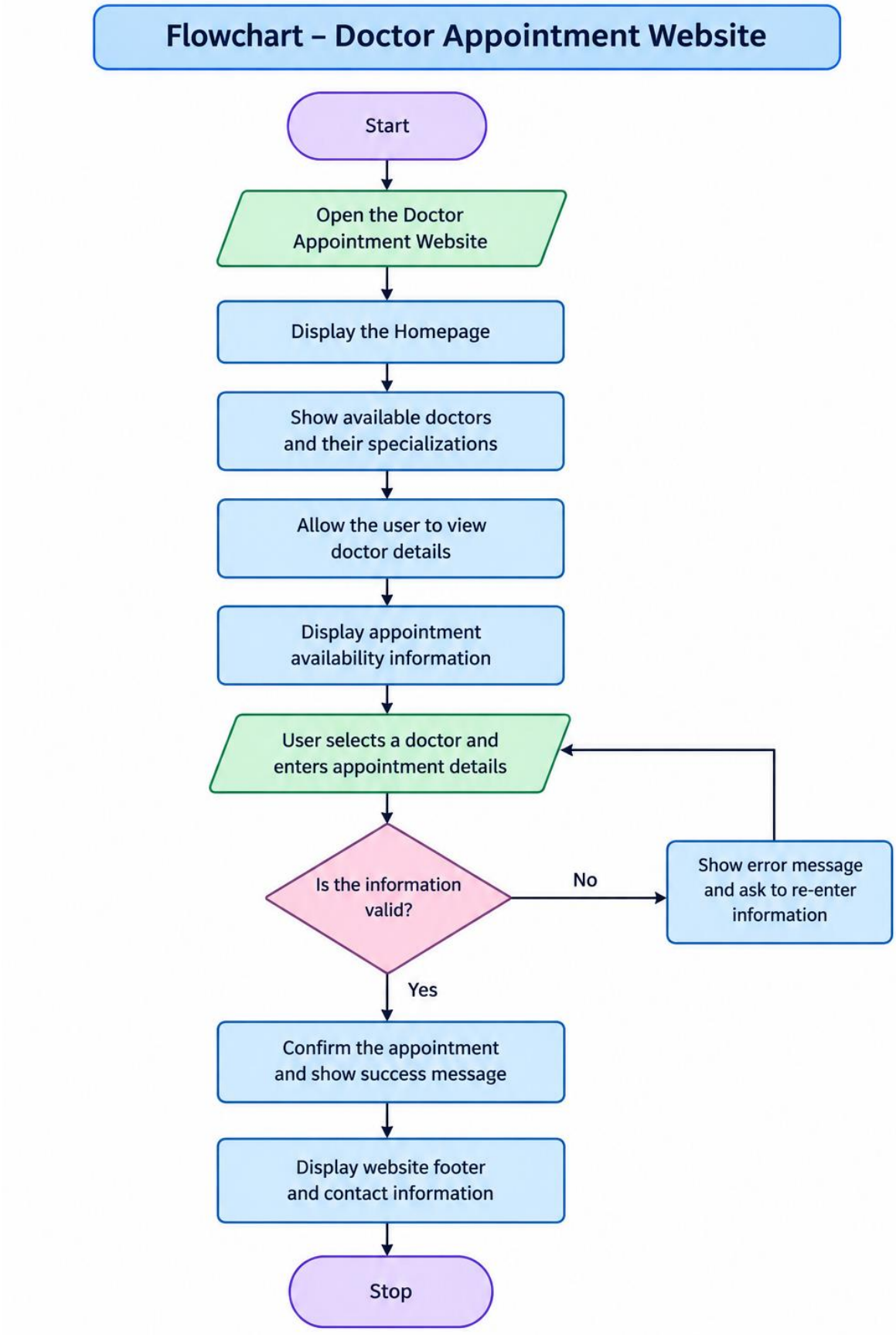
Step 8: Validate the entered information, if applicable.

Step 9: Display the appointment confirmation message, if the booking feature is implemented.

Step 10: Display the website footer and contact information.

Step 11: Stop the process.

5.Flowchart


Flowchart – Doctor Appointment Website
Start
Open the Doctor Appointment Website
Display the Homepage
Show available doctors and their specializations
Allow the user to view doctor details
Display appointment availability information
User selects a doctor and enters appointment details
Is the information valid?
No
Show error message and ask to re-enter information
Yes
Confirm the appointment and show success message
Display website footer and contact information
Stop

6.Pseudocode

BEGIN

1. Start the application.
2. Display the homepage.
3. Display available doctors and their specializations.
4. Display doctor details.
5. Display appointment availability information.
6. Allow users to select a doctor.
7. Allow users to enter appointment details.
8. Display the entered appointment information.
9. Display appointment confirmation information.
10. Display hospital contact information.
11. Apply responsive CSS for different screen sizes.
12. Stop the application.

END

7.HTML/CSS Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CarePlus - Doctor Appointment</title>

    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
```

```
body {
    font-family: Arial, sans-serif;
    background: #f4f9fc;
    color: #263746;
    line-height: 1.6;
}

header {
    background: #087f8c;
    color: white;
    padding: 18px 7%;
    display: flex;
    justify-content: space-between;
    align-items: center;
    flex-wrap: wrap;
    gap: 12px;
}

header h1 {
    font-size: 28px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 10px;
    font-weight: bold;
}
```

```
nav a:hover {
    text-decoration: underline;
}

.hero {
    text-align: center;
    padding: 55px 20px;
    background: linear-gradient(120deg, #087f8c, #38b6a8);
    color: white;
}

.hero h2 {
    font-size: 34px;
    margin-bottom: 12px;
}

.hero p {
    margin-bottom: 20px;
}

.btn {
    display: inline-block;
    background: #087f8c;
    color: white;
    padding: 11px 18px;
    border: none;
    border-radius: 6px;
    text-decoration: none;
```

```
    cursor: pointer;
    font-size: 15px;
}

.hero .btn {
    background: white;
    color: #087f8c;
    font-weight: bold;
}

.btn:hover {
    opacity: 0.85;
}

section {
    padding: 40px 7%;
}

.section-title {
    text-align: center;
    color: #087f8c;
    margin-bottom: 25px;
}

.doctor-container {
    display: grid;
    grid-template-columns:
        repeat(auto-fit, minmax(240px, 1fr));
    gap: 22px;
```

```
}

.doctor-card {
    background: white;
    padding: 24px;
    border-radius: 12px;
    text-align: center;
    box-shadow: 0 4px 14px #00000012;
}

.doctor-icon {
    width: 90px;
    height: 90px;
    margin: 0 auto 12px;
    border-radius: 50%;
    background: #d9f3f2;
    display: grid;
    place-items: center;
    font-size: 48px;
}

.doctor-card h3 {
    color: #087f8c;
}

.doctor-card p {
    margin: 7px 0;
    font-size: 14px;
}
```

```
.specialization {
    display: inline-block;
    background: #e3f7f5;
    padding: 5px 10px;
    border-radius: 20px;
    font-size: 13px;
    color: #087f8c;
    margin: 8px 0;
}

.form-container {
    max-width: 760px;
    margin: auto;
    background: white;
    padding: 28px;
    border-radius: 12px;
    box-shadow: 0 4px 16px #00000012;
}

.form-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 18px;
}

.form-group {
    display: flex;
    flex-direction: column;
```

```
}

.full-width {
    grid-column: 1 / -1;
}

label {
    font-weight: bold;
    margin-bottom: 6px;
    color: #263746;
}

input, select, textarea {
    width: 100%;
    padding: 11px;
    border: 1px solid #bdcbd4;
    border-radius: 6px;
    font-size: 15px;
    font-family: inherit;
}

input:focus, select:focus, textarea:focus {
    outline: 2px solid #38b6a8;
    border-color: #087f8c;
}

textarea {
    resize: vertical;
    min-height: 90px;
```

```
}

.submit-btn {
    width: 100%;
    margin-top: 20px;
    font-weight: bold;
}

#message {
    margin-top: 15px;
    color: #087f55;
    font-weight: bold;
    text-align: center;
}

footer {
    text-align: center;
    padding: 22px;
    background: #075b65;
    color: white;
}

@media (max-width: 600px) {
    header {
        flex-direction: column;
        text-align: center;
    }

    nav a {
```

```
            display: inline-block;
            margin: 5px;
        }

        .hero h2 {
            font-size: 26px;
        }

        .form-grid {
            grid-template-columns: 1fr;
        }

        .full-width {
            grid-column: auto;
        }

        section {
            padding: 30px 5%;
        }

        .form-container {
            padding: 20px;
        }
    }
  </style>
</head>

<body>
```

```
<!-- Navigation -->
<header>
    <h1>CarePlus 🏥</h1>
    <nav>
        <a href="#home">Home</a>
        <a href="#doctors">Doctors</a>
        <a href="#appointment">Book Appointment</a>
    </nav>
</header>

<!-- Hero Section -->
<section class="hero" id="home">
    <h2>Your Health, Our Priority</h2>
    <p>Find the right doctor and schedule your consultation easily.</p>
    <a href="#appointment" class="btn">Book an Appointment</a>
</section>

<!-- Doctor Profiles -->
<section id="doctors">
    <h2 class="section-title">Our Doctors</h2>

    <div class="doctor-container">

        <div class="doctor-card">
            <div class="doctor-icon">👨‍⚕️</div>
            <h3>Dr. Rahul Sharma</h3>
            <span class="specialization">Cardiologist</span>
            <p>Experience: 10 years</p>
            <p>Consultation: 9:00 AM - 12:00 PM</p>
```

```
    <p>Available slots: 9 AM, 10 AM, 11 AM</p>
    <button class="btn"
        onclick="selectDoctor('Dr. Rahul Sharma')">
        Select Doctor
    </button>
</div>

<div class="doctor-card">
    <div class="doctor-icon">👩□</div>
    <h3>Dr. Priya Reddy</h3>
    <span class="specialization">Dermatologist</span>
    <p>Experience: 8 years</p>
    <p>Consultation: 10:00 AM - 1:00 PM</p>
    <p>Available slots: 10 AM, 11 AM, 12 PM</p>
    <button class="btn"
        onclick="selectDoctor('Dr. Priya Reddy')">
        Select Doctor
    </button>
</div>

<div class="doctor-card">
    <div class="doctor-icon">👨□</div>
    <h3>Dr. Arjun Kumar</h3>
    <span class="specialization">Pediatrician</span>
    <p>Experience: 7 years</p>
    <p>Consultation: 2:00 PM - 5:00 PM</p>
    <p>Available slots: 2 PM, 3 PM, 4 PM</p>
    <button class="btn"
        onclick="selectDoctor('Dr. Arjun Kumar')">
```

```
            Select Doctor
        </button>
    </div>

</div>
</section>

<!-- Patient Registration and Appointment Form -->
<section id="appointment">
    <h2 class="section-title">Patient Registration & Appointment</h2>

    <div class="form-container">
        <form id="appointmentForm">

            <div class="form-grid">

                <div class="form-group">
                    <label for="patientName">Patient Full Name</label>
                    <input type="text" id="patientName"
                        placeholder="Enter your full name"
                        autocomplete="name" required>
                </div>

                <div class="form-group">
                    <label for="age">Age</label>
                    <input type="number" id="age"
                        min="1" max="120"
                        placeholder="Enter your age" required>
                </div>
```

```
<div class="form-group">
    <label for="gender">Gender</label>
    <select id="gender" required>
        <option value="">Select gender</option>
        <option>Female</option>
        <option>Male</option>
        <option>Prefer not to say</option>
    </select>
</div>

<div class="form-group">
    <label for="phone">Mobile Number</label>
    <input type="tel" id="phone"
        placeholder="Enter mobile number"
        autocomplete="tel"
        pattern="[0-9]{10}"
        maxlength="10"
        title="Enter a 10-digit mobile number"
        required>
</div>

<div class="form-group">
    <label for="email">Email Address</label>
    <input type="email" id="email"
        placeholder="Enter your email"
        autocomplete="email" required>
</div>
```

```
<div class="form-group">
    <label for="date">Appointment Date</label>
    <input type="date" id="date" required>
</div>

<div class="form-group">
    <label for="doctor">Choose Doctor</label>
    <select id="doctor" required>
        <option value="">Select a doctor</option>
        <option value="Dr. Rahul Sharma">
            Dr. Rahul Sharma - Cardiologist
        </option>
        <option value="Dr. Priya Reddy">
            Dr. Priya Reddy - Dermatologist
        </option>
        <option value="Dr. Arjun Kumar">
            Dr. Arjun Kumar - Pediatrician
        </option>
    </select>
</div>

<div class="form-group">
    <label for="slot">Available Appointment Slot</label>
    <select id="slot" required>
        <option value="">Choose doctor first</option>
    </select>
</div>

<div class="form-group full-width">
```

```
                <label for="reason">Reason for Consultation</label>
                <textarea id="reason"
                    placeholder="Briefly describe the reason for your visit"
                    required></textarea>
            </div>

        </div>

        <button type="submit" class="btn submit-btn">
            Confirm Appointment
        </button>

        <p id="message" role="status" aria-live="polite"></p>

    </form>
  </div>
</section>

<!-- Footer -->
<footer>
  <p>&copy; 2026 CarePlus Clinic. All rights reserved.</p>
  <p>Your health matters to us.</p>
</footer>

<script>
  const doctorSlots = {
    "Dr. Rahul Sharma": ["9:00 AM", "10:00 AM", "11:00 AM"],
    "Dr. Priya Reddy": ["10:00 AM", "11:00 AM", "12:00 PM"],
    "Dr. Arjun Kumar": ["2:00 PM", "3:00 PM", "4:00 PM"]
```

```
};

const doctorSelect = document.getElementById("doctor");
const slotSelect = document.getElementById("slot");
const dateInput = document.getElementById("date");
const message = document.getElementById("message");

// Prevent selecting a date in the past.
const today = new Date();
const localToday = new Date(
    today.getTime() - today.getTimezoneOffset() * 60000
).toISOString().split("T")[0];

dateInput.min = localToday;

function selectDoctor(name) {
    doctorSelect.value = name;
    updateSlots();
    document.getElementById("appointment").scrollIntoView({
        behavior: "smooth"
    });
}

function updateSlots() {
    const doctor = doctorSelect.value;
    slotSelect.innerHTML = "";

    if (!doctor || !doctorSlots[doctor]) {
        slotSelect.add(new Option("Choose doctor first", ""));
```

```
        return;
    }

    slotSelect.add(new Option("Select a time slot", ""));

    doctorSlots[doctor].forEach(function(slot) {
        slotSelect.add(new Option(slot, slot));
    });
}

doctorSelect.addEventListener("change", updateSlots);

document.getElementById("appointmentForm")
    .addEventListener("submit", function(event) {
        event.preventDefault();

        if (dateInput.value < localToday) {
            message.textContent =
                "Please select today or a future date.";
            return;
        }

        const name =
            document.getElementById("patientName").value.trim();
        const doctor = doctorSelect.value;
        const date = dateInput.value;
        const slot = slotSelect.value;

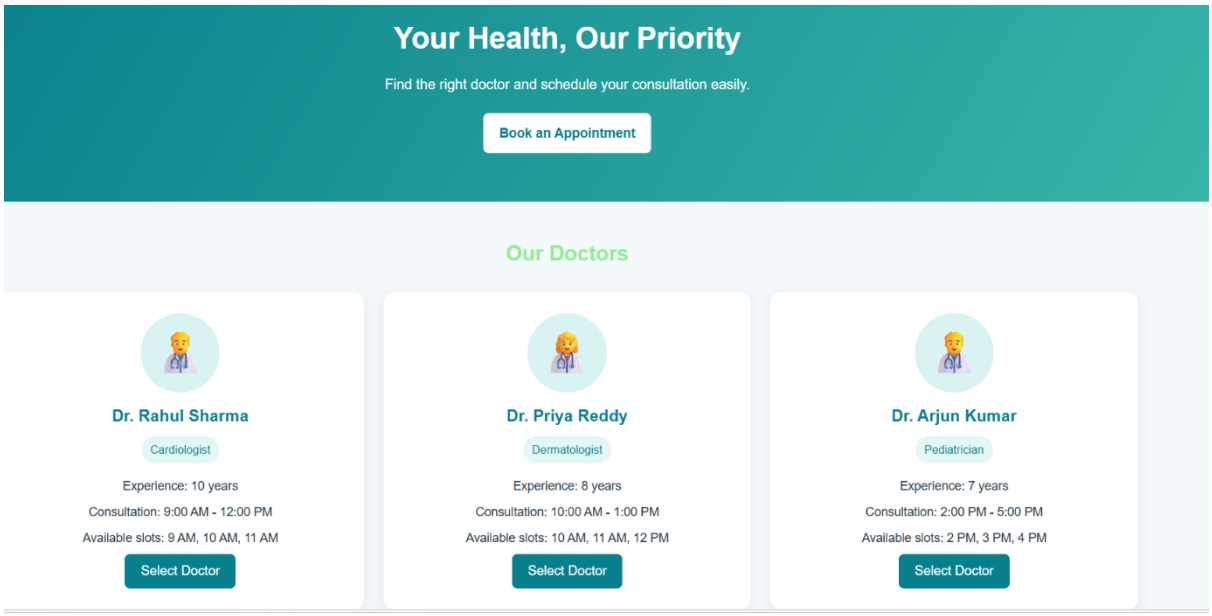
        message.textContent =
```

```
            "Thank you, " + name + "! Your appointment request with " +
            doctor + " for " + date + " at " + slot +
            " has been recorded in this demo.";

        this.reset();
        slotSelect.innerHTML =
            '<option value="">Choose doctor first</option>';
        dateInput.min = localToday;
    });
</script>

</body>
</html>
```

7.Output Screenshots



Patient Registration & Appointment

Patient Full Name

Enter your full name

Age

Enter your age

Gender

Select gender

Mobile Number

Enter mobile number

Email Address

Enter your email

Appointment Date

dd-mm-yyyy

Choose Doctor

Select a doctor

Available Appointment Slot

Choose doctor first

Reason for Consultation

Briefly describe the reason for your visit

Confirm Appointment